

Helix OTA — Continuation

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1. Current state

Field	Value
HEAD	998c6af2 chore(workable-now): full verification sweep – gofmt, §11.4.166 semgrep, A/B re-proof, recordings, Status reconcile
Phase	Stable / release-readiness — all autonomous work GREEN; remaining items hardware/operator-gated
Terminal goal	Fully validated Helix OTA control plane driving real Android A/B updates end-to-end (protocol round-trip → payload apply → slot switch → rollback) on emulated + physical targets

Latest session (2026-06-22) — “do everything workable” sweep

All autonomous items GREEN with real captured evidence (no bluffs): - Test sweep GREEN (server 11/11 + 4 Go submodules 3/3, –race 0); **gofmt fixed** (13 files); pre-build gates PASS; docs_chain in-sync. - **Local QEMU A/B OTA 6/6 GREEN** (smoke/boot/slot-switch/rollback/OTA-apply) — evidence docs/qa/20260622T07*. Fixed a real §11.4.115 RED-mode polarity isolation bug in ab_rauc_verity.sh (RED×2+GREEN×2 deterministic). - **10 server-feature recordings regenerated**, vision-verified (203 HTTP assertions) — docs/qa/20260622-server-recordings-regen/; Status.md §11.4.153 reconciled to durable paths. - **§11.4.166 Semgrep**: propagated §11.4.160-166 into CLAUDE/AGENTS/GEMINI; tokenless local scan wired into commit_all.sh; constitution pin → 09d8940 (adds docs/semgrep/TOKEN_SETUP.md); 7 findings triaged → **0 remaining** (TLS MinVersion fix + cited suppressions). - Submodule push parity closed (ota-update-engine-bridge, ota-android-agent); GitFlic “blocker” disproven; §11.4.30 transient qa-results/ untracked+ignored (232 files). - nezha AVD boot proven; real Android A/B = honest **operator-attended SKIP** (Cuttlefish unprovisioned — needs sudo+reboot+fetch_cvd).

Operator action items: (1) Semgrep SEMGREP_APP_TOKEN — follow `constitution/docs/semgrep/TOKEN_SETUP.md` (`semgrep login`) to silence the MCP hook (optional; tokenless gate already compliant); (2) provision Cuttlefish on nezha to unblock real Android A/B; (3) RK3588 board for OTA-004/F55/F56.

Active items

ID	Title	Status	Type
OTA-003	Emulator Tier-2 — real Android A/ B (update_engine/ AVB/dm-verity auto-rollback)	In testing	Task
OTA-004	Emulator Tier-3 — real RK3588 / Orange Pi 5 Max vendor HAL, U- Boot slot-switch, dm-verity on real partitions	Operator-blocked	Task

Recently closed items

ID	Title	Status
OTA-021	HelixTrack bidirectional sync verification	Completed
OTA-020	Database migration test	Fixed
OTA-019	Build resource stats tracker	Completed
OTA-018	ApplyPort CLI + slot manager + Ed25519 verifier	Implemented
OTA-017	U-Boot corrupt-slot auto- rollback via bootcount	Implemented
OTA-016	RAUC dd-apply to inactive slot with dm- verity	Implemented
OTA-015	A/B slot switch via U- Boot BOOT_ORDER	Implemented

Full issue details: `docs/Issues.md` · `docs/Fixed.md` · `docs/Issues_Summary.md` · `docs/Fixed_Summary.md`.

2. Server tests — all GREEN

All 13 testable packages pass (11 internal packages + chaos + stress suites):

```
ok internal/api
ok internal/config
ok internal/device
ok internal/deviceemu
ok internal/fabric
ok internal/health
ok internal/rollout
ok internal/store
ok internal/transport
ok tests/chaos
ok tests/stress
```

No-test-file packages (build-only): cmd/applyport, cmd/ota-device-emu, cmd/ota-server, tools/loadtest.

3. Emulator status

Property	Value
Host	nezha.local — Linux x86_64, 62 GB RAM, KVM
Image	Android API 36 (Android 16) — CZ_API36_Phone
Transport	ADB over SSH tunnel (reachable at emulator-5554 from nezha)
ADB local	No Android devices connected locally (emulator is remote via SSH tunnel)
LD_LIBRARY_PATH	/home/milosvasic/.local/lib
Cuttlefish Tier-2	Pending AOSP guest images

The HelixTrack API is accessible from the emulator via SSH tunnel. The CZ_API36_Phone image provides the standard Android 16 AOSP stack on an x86_64 KVM host.

4. What's running

- **Main stream:** Emulator Tier-2 validation (remote on nezha.local)
 - **Background agents:** None currently dispatched
 - **Recording directory:** \$HOME/Downloads per §11.4.158(D) (default; no project-level override)
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5. Next actions (priority-ordered)

1. **OTA-003 — Emulator Tier-2 inline testing.** Verify the Android emulator on nezha.local can register with the control plane, receive an OTA payload, apply it

- via `update_engine`, A/B slot-switch, and auto-rollback on corruption. Drive end-to-end through the real user-equivalent path per §11.4.143.
2. **OTA-004 — Hardware unblock.** When a physical RK3588 / Orange Pi 5 Max board becomes reachable over ADB/SSH, flash and validate Tier-3 (vendor HAL, U-Boot slot-switch, real-partition dm-verity).
 3. **Feature-coverage video recording.** Produce §11.4.153 mandatory per-feature real-use videos confirming every server endpoint, every emulator tier, and every submodule works — with §11.4.159 window-scoped MP4 capture + §11.4.160 vision verification.
 4. **Standing regression-guard suite (§11.4.135).** Ensure every closed OTA-NNN item has its §11.4.115 polarity-switch regression test registered in the suite.
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6. Binding constraints

- **Anti-bluff (§11.4 / §107):** Every PASS carries captured physical evidence per §11.4.5 / §11.4.69 / §11.4.107. Metadata-only / config-only / absence-of-error / grep-without-runtime PASS are all forbidden.
 - **No-force-push (§11.4.113):** `git push --force`, `--force-with-lease`, `+<ref>` are STRICTLY FORBIDDEN. Always merge onto latest main and push fast-forward-only.
 - **Commit via `commit_all.sh`:** All changes use the project's canonical commit wrapper. Direct `git add/git commit/git push` are never used.
 - **Four-layer fix-verification (§11.4.108):** SOURCE → ARTIFACT → RUNTIME-ON-CLEAN-TARGET → USER-VISIBLE. Runtime signature is the definition of done.
 - **Independent review (§11.4.142 / §11.4.125):** Every change passes an independent code-review agent before build. Review iterates to zero-finding GO per §11.4.134.
 - **Multi-upstream push (§2.1):** Every commit fans out to all four upstreams (GitHub + GitLab + GitFlic + GitVerse).
 - **Rootless containers (§11.4.161):** All containerized workloads use Podman in rootless mode via `vasic-digital/containers` submodule.
 - **No remote CI (§11.4.156):** All CI/CD automation is DISABLED. Enforcement is through local git hooks + `pre_build_verification.sh`.
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7. Feature tracking

Feature inventory and per-row status: `docs/features/Status.md` (Rev 11, 2026-06-21). Summary companion: `docs/features/Status_Summary.md`.

8. Fresh session start

```
git fetch --all --prune --tags
```

Then read this file (docs/CONTINUATION.md) and docs/Issues.md for the full active-item context. The single highest-priority next action is **Emulator Tier-2 inline testing on nezha.local** (OTA-003) — drive the Android emulator through the full OTA lifecycle against the control plane.